

Extension of Conformal Prediction

Conformal Risk Control

$Y_{n+1} \in \mathcal{C}(X_{n+1})$ only describes one kind of risks, for some other tasks, other notions of error can be more helpful:

- **Coordinatewise coverage rate.** If $Y_{n+1} \in \mathbb{R}^d$, we may want to return a prediction set such that most coordinates of the response Y_{n+1} are contained in their respective prediction intervals, i.e., we want the coordinatewise miscoverage rate

$$\frac{1}{d} \sum_{j=1}^d \mathbb{1} \{ (Y_{n+1})_j \notin \mathcal{C}(X_{n+1})_j \}$$

to be bounded in expectation by some target error level α , where our prediction set is of the form $\mathcal{C}(X_{n+1}) = \mathcal{C}(X_{n+1})_1 \times \cdots \times \mathcal{C}(X_{n+1})_d$.

- **Accuracy in hierarchical classification.** In hierarchical classification, the classes $\mathcal{Y}$ are the leaf nodes of a hierarchy—for example, a ‘poodle’ is an instance of a ‘dog’, which is an instance of an ‘animal’, and so on. We consider making predictions that correspond to nodes in the hierarchy—predicting a leaf node (‘poodle’) is a high-resolution prediction, whereas an interior node (‘animal’) is a low-resolution prediction. Here, we wish to bound the probability of event that we have made an error,

$$\mathbb{1} \{ \hat{Y}_{n+1} \text{ is not an ancestor of } Y_{n+1} \text{ in the hierarchy} \},$$

by some target level α .

- **False negative rate.** Consider a setting where $Y_{n+1} \in \{0, 1\}^d$, with a 1 indicating the presence of some signal (e.g., the j th pixel of the image contains a vehicle). Then we might output a vector $\hat{Y}_{n+1} \in \{0, 1\}^d$ indicating where we believe the true signals lie, and would like to require that our set of (estimated) 1's contains most of the positives. Here, we would aim to control the false negative rate:

$$\frac{\sum_{j=1}^d \mathbb{1} \{ (Y_{n+1})_j = 1, (\hat{Y}_{n+1})_j = 0 \}}{\sum_{j=1}^d \mathbb{1} \{ (Y_{n+1})_j = 1 \}}$$

to be bounded (in expectation) by some target level α .

Note that marginal predictive coverage can also be expressed as control of the expectation of miscoverage indicator,

$$\mathbb{1}\{Y_{n+1} \notin \mathcal{C}(X_{n+1})\}. \quad (10.1)$$

Thus, marginal miscoverage, and the three other examples above, can all be thought of as a *risk*—the expected value of a loss. The basic idea of conformal risk control is to replace the indicator of miscoverage with other loss functions, to generalize beyond the goal of marginal coverage. Below we present the extension to arbitrary monotone loss functions.

We now introduce our notation. Let $\mathcal{C}_\lambda$ denote the set-valued output of a procedure, with some tuning parameter λ —for instance, $\mathcal{C}_\lambda(X)$ might denote a prediction interval for the response Y . The tuning parameter $\lambda \in \Lambda \subseteq \mathbb{R}$ indexes a nested family of sets, with

$$\lambda_1 \leq \lambda_2 \implies \mathcal{C}_{\lambda_1}(x) \subseteq \mathcal{C}_{\lambda_2}(x). \quad (10.2)$$

In particular, split conformal prediction with a pretrained score function s corresponds to choosing $\mathcal{C}_\lambda(x) = \{y \in \mathcal{Y} : s(x, y) \leq \lambda\}$.

As our notion of statistical error, let L be a *loss function* on prediction sets, which is assumed to be monotone in $\mathcal{C}$,

$$\mathcal{C} \subseteq \mathcal{C}' \implies L(y, \mathcal{C}) \geq L(y, \mathcal{C}'). \quad (10.3)$$

i.e., larger sets lead to smaller loss. For instance, for conformal prediction we have the miscoverage loss $L(y, \mathcal{C}) = \mathbb{1}\{y \notin \mathcal{C}\}$ (as in (10.1)), but this notation can accommodate more general settings such as the examples above. We hope to choose a value of λ so that the confidence set $\mathcal{C}_\lambda$ satisfies

$$\mathbb{E}[L(Y_{n+1}, \mathcal{C}_\lambda(X_{n+1}))] \leq \alpha, \quad (10.4)$$

a property we refer to as *risk control*.

over $X_{n+1}, Y_{n+1} \sim D_n$

$$L(Y_{n+1}, \mathcal{C}_\lambda(X_{n+1})) = \mathbb{1}(Y_{n+1} \notin \mathcal{C}_\lambda(X_{n+1}))$$

Calibration data.

As hinted above, we can extend the split conformal prediction algorithm in this more general setting. The algorithm is as follows. Consider a plug-in estimator of the expected loss at each value of the tuning parameter $\lambda \in \Lambda$, given by

$$\hat{R}(\lambda) = \frac{1}{n} \sum_{i=1}^n L(Y_i, \mathcal{C}_\lambda(X_i)). \leq \alpha$$

$\alpha' = \alpha - \frac{1-\alpha}{n}$

Intuitively, we would expect that we can then choose a value of λ satisfying $\hat{R}(\lambda) \leq \alpha$ to achieve the desired bound on expected loss, but this does not lead to risk control due to finite-sample fluctuations in $\hat{R}(\lambda)$. To adjust for this and obtain the guarantee in (10.4), our procedure needs to be slightly more conservative. For a loss L that takes values in $[0, 1]$, we select a value of λ where the plug-in estimate of risk is slightly smaller than α :

$$\hat{\lambda} = \inf \left\{ \lambda : \hat{R}(\lambda) \leq \underbrace{\alpha - (1-\alpha)/n}_{\alpha'} \right\}. \quad (10.5)$$

$\mathcal{C}_\lambda$

The next result shows that this procedure provides risk control. For simplicity, we will assume $\Lambda = \mathbb{R}$. In order to ensure that $\hat{\lambda}$ is well-defined, we will assume that $\alpha - (1-\alpha)/n > 0$, and that for any (x, y) , it holds that $\lim_{\lambda \rightarrow \infty} L(y, \mathcal{C}_\lambda(x)) = 0$ and $\lim_{\lambda \rightarrow -\infty} L(y, \mathcal{C}_\lambda(x)) = 1$.

bounded

Theorem 10.1: Conformal risk control

Suppose that $(X_1, Y_1), \dots, (X_{n+1}, Y_{n+1})$ are exchangeable, and that the loss L takes values in $[0, 1]$. Assume that the map $\lambda \mapsto L(y, \mathcal{C}_\lambda(x))$ is right-continuous and is monotone nonincreasing, for any (x, y) . Then, with $\hat{\lambda}$ selected as in (10.5), we have

$$\mathbb{E} [L(Y_{n+1}, \mathcal{C}_{\hat{\lambda}}(X_{n+1}))] \leq \alpha.$$

Here, the expectation is over both the calibration data (i.e., the randomness in $\hat{\lambda}$) and the test point, as with conformal prediction. Note that the monotonicity condition (i.e., requiring $\lambda \mapsto L(y, \mathcal{C}_\lambda(x))$ to be nonincreasing) is an immediate consequence of assuming nested sets $\mathcal{C}_\lambda$, as in (10.2), and monotonicity of the loss L , as in (10.3). The assumption of right-continuity holds in many practical instances, such as split conformal and the other examples above.

Proof of Theorem 10.1

Step 1: Deterministic inequalities on monotone risks. Consider an arbitrary dataset $\mathcal{D} = ((x_1, y_1), \dots, (x_m, y_m))$. We will denote the empirical risk on the dataset as

$$\hat{R}(\lambda; \mathcal{D}) = \frac{1}{m} \sum_{i=1}^m L(y_i, C_\lambda(x_i))$$

and define the following family of thresholds:

$$\hat{\lambda}(\mathcal{D}, \beta) = \inf \left\{ \lambda : \hat{R}(\lambda; \mathcal{D}) \leq \beta \right\},$$

for $\beta \in (0, 1)$. The right-continuity of L implies that $\hat{R}(\lambda; \mathcal{D})$ is right-continuous for

$\hat{\lambda}$ used for CRC

any $\mathcal{D}$, giving us the following deterministic inequality for any β :

$$\hat{R}(\hat{\lambda}(\mathcal{D}; \beta); \mathcal{D}) \leq \beta. \quad (10.6)$$

Step 2: Applying exchangeability. The family $\hat{\lambda}(\mathcal{D}, \beta)$, defined above, generalizes our previous choice of threshold in (10.5): we have that $\hat{\lambda} = \hat{\lambda}(\mathcal{D}_n, \alpha')$, where $\mathcal{D}_n = ((X_1, Y_1), \dots, (X_n, Y_n))$ and $\alpha' = \alpha - (1 - \alpha)/n$. However, here we will consider applying this definition to a different dataset—not to the calibration set $\mathcal{D}_n$, but instead, to the dataset $\mathcal{D}_{n+1} = ((X_i, Y_i))_{i \in [n+1]}$ containing both the calibration set and test point: define

$$\lambda^* = \hat{\lambda}(\mathcal{D}_{n+1}, \alpha)$$

$$C_{\lambda^*} \rightarrow \mathbb{E}[L(Y_{n+1}, C_{\lambda^*})] \leq \alpha$$

(note that here we use the original target risk level α , not the modified level α'). Since the data points are exchangeable, and λ^* is a symmetric function of the dataset $\mathcal{D}_{n+1}$, we have

$$\mathbb{E}[L(Y_{n+1}, C_{\lambda^*}(X_{n+1}))] = \mathbb{E}[L(Y_i, C_{\lambda^*}(X_i))],$$

for all $i \in [n+1]$ (since, by Lemma 2.3, the data points $(X_1, Y_1), \dots, (X_{n+1}, Y_{n+1})$ are exchangeable when we condition on λ^* , and in particular this implies $L(Y_{n+1}, C_{\lambda^*}(X_{n+1})) \stackrel{d}{=} L(Y_i, C_{\lambda^*}(X_i))$). Therefore,

$$\mathbb{E}[L(Y_{n+1}, C_{\lambda^*}(X_{n+1}))] = \frac{1}{n+1} \sum_{i=1}^{n+1} \mathbb{E}[L(Y_i, C_{\lambda^*}(X_i))] \stackrel{\text{def}}{=} \mathbb{E}[\hat{R}(\lambda^*; \mathcal{D}_{n+1})] \leq \alpha,$$

where the first step holds by exchangeability, the second step by definition of $\hat{R}$, and the third step by (10.6).

Step 3: proving the bound. Since we have assumed L takes values in $[0, 1]$, we have

$$\hat{R}(\lambda; \mathcal{D}_{n+1}) \leq \frac{1}{n+1} + \frac{n}{n+1} \hat{R}(\lambda; \mathcal{D}_n) \leq \frac{1}{n+1} + \frac{n}{n+1} (\alpha - \frac{1-\alpha}{n}) \leq \alpha$$

Therefore, for any λ , by definition of α' we have

$$\hat{\lambda} = \inf \{ \lambda : \hat{R}(\lambda; \mathcal{D}_n) \leq \alpha' \} \Rightarrow \hat{R}(\lambda; \mathcal{D}_{n+1}) \leq \alpha : \lambda \} \inf = \lambda^*$$

and so it must hold that $\lambda^* \leq \hat{\lambda}$. Monotonicity of L implies that $L(Y_{n+1}, C_{\hat{\lambda}}(X_{n+1})) \leq L(Y_{n+1}, C_{\lambda^*}(X_{n+1}))$ holds surely, and therefore

$$\mathbb{E}[L(Y_{n+1}, C_{\hat{\lambda}}(X_{n+1}))] \leq \mathbb{E}[L(Y_{n+1}, C_{\lambda^*}(X_{n+1}))] \leq \alpha.$$

$$\lambda^* \leq \hat{\lambda} \Rightarrow \mathbb{E}[L(Y_{n+1}, C_{\hat{\lambda}}(X_{n+1}))] \leq \mathbb{E}[L(Y_{n+1}, C_{\lambda^*}(X_{n+1}))] \leq \alpha$$

obtained CRC.

$X_1, \dots, X_n$ are exch.

$f(X_1, \dots, X_n)$ is sym.

$X_1, \dots, X_n \mid f$ are exch.

λ is sym.

$(X_1, \xi_1) \dots (X_{n+1}, \xi_{n+1}) \mid \lambda$ is exch.
 $\downarrow$
 C_λ

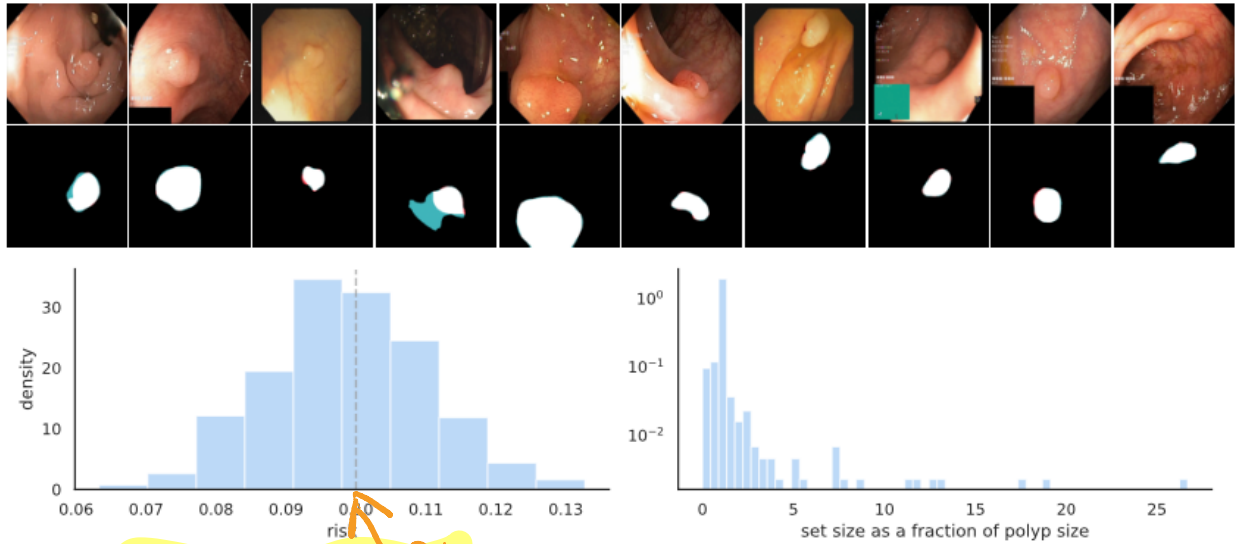


Figure 1: **FNR control in tumor segmentation**. The top figure shows examples of our procedure with correct pixels in white, false positives in blue, and false negatives in red. The bottom plots report FNR and set size over 1000 independent random data splits. The dashed gray line marks α .

In tumor segmentation, our input is a $d \times d$ image and our label is a set of pixels $Y_i \in \wp(\{(1, 1), (1, 2), \dots, (d, d)\})$, with $\wp$ the power set. We use an image segmentation model $f : \mathcal{X} \rightarrow [0, 1]^{d \times d}$ outputting a probability for each pixel and measure loss as the fraction of false negatives,

$$L_i^{\text{FNR}}(\lambda) = 1 - \frac{|Y_i \cap \mathcal{C}_\lambda(X_i)|}{|Y_i|}, \text{ where } \mathcal{C}_\lambda(X_i) = \{y : f(X_i)_y \geq 1 - \lambda\}. \quad (7)$$

The expected value of L_i^{FNR} is the FNR. Since L_i^{FNR} is monotone, so is the FNR. Thus, we use the technique in Section 2.1 to pick $\hat{\lambda}$ by (4) that controls the FNR on a new point, which guarantees:

$$\mathbb{E}[L_{n+1}^{\text{FNR}}(\hat{\lambda})] \leq \alpha. \quad (8)$$

$$\mathbb{E}[\text{FNR}] \leq \alpha.$$

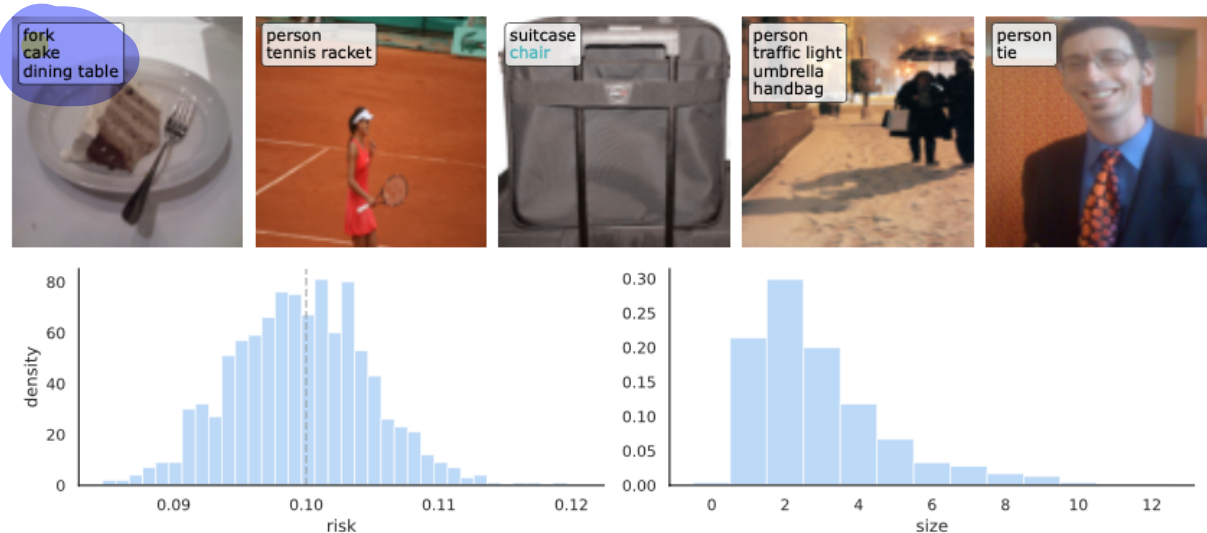


Figure 2: **FNR control on MS COCO**. The top figure shows examples of our procedure with correct classes in black, false positives in blue, and false negatives in red. The bottom plots report FNR and set size over 1000 independent random data splits. The dashed gray line marks α .

3.2 FNR CONTROL IN MULTILABEL CLASSIFICATION

In the multilabel classification setting, our input X_i is an image and our label is a set of classes $Y_i \subset \{1, \dots, K\}$ for some number of classes K . Using a multiclass classification model $f : \mathcal{X} \rightarrow [0, 1]^K$, we form prediction sets and calculate the number of false positives exactly as in (7). By Theorem 1, picking $\hat{\lambda}$ as in (4) again yields the FNR-control guarantee in (8). We evaluate on the Microsoft Common Objects in Context (MS COCO) dataset (Lin et al., 2014), a large-scale 80-class multiclass classification task commonly used in computer vision. We choose a TResNet (Ridnik et al., 2020) as our model f and used $n = 4000$, and evaluated risk control with 1000 validation data points. We report results with $\alpha = 0.1$ in Figure 2. The mean and standard deviation of the risk over 1000 trials are 0.0996 and 0.0052, respectively. The results indicate that the risk is almost exactly controlled, the spread is not too wide, and the set sizes are reasonable, not overly inflated.

$$P(Y_{n+1} \in C(X_{n+1})) \geq 1 - \alpha$$

$$\mathbb{E}[L(Y_{n+1} - C_\lambda(X_{n+1}))] \leq \alpha$$

individual error

One test data

$$f: \underbrace{X_{n+1}}_{\text{mol feature.}} \rightarrow \underbrace{\hat{Y}_{n+1}}_{\text{chemical property.}}$$

$$D_{\text{test}}: \{X_i, Y_i\}_{i=1}^m$$

Family-wise error rate for multiple test points

Conformal prediction is typically presented as a method for constructing a prediction interval for a single test point X_{n+1} , given a training set of size n . However, it is often of interest to provide prediction intervals for multiple test points in a batch. This is a question of multiplicity: do we need to adjust our construction of the prediction sets to account for the fact that we are performing inference for multiple test points?

Throughout this section, we will use our previous notation for the split conformal prediction setting: we write $\mathcal{D}_n = ((X_i, Y_i))_{i \in [n]}$ to denote the calibration set, while $s : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ is the pretrained score function. Given a set of test data points $X'_1, \dots, X'_m$, we might be interested in

m test pt.

obtaining a guarantee of the form

$$\mathbb{P}(Y'_1 \in \mathcal{C}(X'_1), \dots, Y'_m \in \mathcal{C}(X'_m)) \geq 1 - \alpha_{\text{FWER}}, \quad (10.7)$$

meaning *all* of the m prediction sets cover (with probability at least $1 - \alpha_{\text{FWER}}$). Here α_{FWER} is a bound on the family-wise error rate: the probability of *at least one* miscoverage among our m prediction sets.

$$\mathbb{P}(\text{at least one } Y'_i \notin \mathcal{C}(X'_i)) \leq \alpha_{\text{FWER}}$$

Let us now consider how the split conformal prediction method would perform relative to this goal. Since the prediction set is given by $\mathcal{C}(x) = \{y \in \mathcal{Y} : s(x, y) \leq \hat{q}\}$, where $\hat{q} = \text{Quantile}(S_1, \dots, S_n; (1 - \alpha)(1 + 1/n))$, we can rephrase the goal (10.7) above as

$$\mathbb{P}(S'_1 \leq \hat{q}) \times \dots \times \mathbb{P}(S'_m \leq \hat{q}) \neq \mathbb{P}(S'_1 \leq \hat{q}, \dots, S'_m \leq \hat{q}) \geq 1 - \alpha_{\text{FWER}},$$

where the test point scores are given by $S'_i = s(X'_i, Y'_i)$. The key challenge is that the events $\{S'_i \leq \hat{q}\}_{i \in [m]}$ are not independent, since the conformal quantile $\hat{q}$ (which is random) provides a threshold that is shared across all m events.

In this next result, we consider both the finite-sample setting, where the number of calibration points n is fixed, and an asymptotic regime, where the number of calibration points is $n \rightarrow \infty$ (while the number of test points m , and the pretrained score function s , remain fixed). To accommodate both of these settings, we will write $\hat{q}_n$ (rather than $\hat{q}$) to emphasize that n may be increasing.

size of $\mathcal{D}_{\text{cal}}$

i.i.d. $n \rightarrow \infty$

Proposition 10.2: The FWER of split conformal prediction (asymptotic)

Let $(X'_1, Y'_1), \dots, (X'_m, Y'_m), (X_1, Y_1), (X_2, Y_2), \dots$ be sampled i.i.d. from some distribution P . Let $s: \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ be a pretrained score function, and for each calibration set size $n \geq 1$, define the split conformal prediction set as

sample

$$\mathcal{C}_{\hat{q}_n}(x) = \{y : s(x, y) \leq \hat{q}_n\}$$

population

where

$$\hat{q}_n = \text{Quantile}(S_1, \dots, S_n; (1 - \alpha)(1 + 1/n))$$

for scores $S_i = s(X_i, Y_i)$, $i \in [n]$. Define $q^* = \text{Quantile}(F, 1 - \alpha)$, where F is the CDF of $s(X, Y)$ when $(X, Y) \sim P$.

population

Then for any fixed $n, m \geq 1$,

$$\mathbb{P}(Y'_1 \in \mathcal{C}_{\hat{q}_n}(X'_1), \dots, Y'_m \in \mathcal{C}_{\hat{q}_n}(X'_m)) \geq (1 - \alpha)^m.$$

Moreover, if $F(q)$ is continuous and strictly increasing at $q = q^*$, then for any fixed $m \geq 1$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(Y'_1 \in \mathcal{C}_{\hat{q}_n}(X'_1), \dots, Y'_m \in \mathcal{C}_{\hat{q}_n}(X'_m)) = (1 - \alpha)^m.$$

$0.9^m \approx 0$ when m is large

Proof of Proposition 10.2

By construction of the split conformal prediction set, $Y'_i \in \mathcal{C}_{\hat{q}_n}(X'_i)$ holds if and only if $s(X'_i, Y'_i) \leq \hat{q}_n$. Since the test point scores $S'_i = s(X'_i, Y'_i)$ are i.i.d. draws from the CDF F , and $\hat{q}_n$ depends only on the calibration data (and is therefore independent of the test scores), we have

$$\mathbb{P}(S'_1 \leq \hat{q}_n, \dots, S'_m \leq \hat{q}_n) = \mathbb{E}[\mathbb{P}(S'_1 \leq \hat{q}_n, \dots, S'_m \leq \hat{q}_n \mid \hat{q}_n)] = \mathbb{E}[F(\hat{q}_n)^m] = \{\mathbb{E}[F(\hat{q}_n)]\}^m$$

The function $t \mapsto t^m$ is convex on $t \geq 0$, so $\mathbb{E}[F(\hat{q}_n)^m] \geq \mathbb{E}[F(\hat{q}_n)]^m$ by Jensen's inequality. Finally, by definition of F , we can write $F(\hat{q}_n) = \mathbb{P}(s(X'_1, Y'_1) \leq \hat{q}_n \mid \hat{q}_n) =$

$\hat{q}_n$ from $\mathcal{D}_{cal}$

$$(1 - \alpha)^m$$

$\mathbb{P}(Y'_1 \in \mathcal{C}_{\hat{q}_n}(X'_1) \mid \hat{q}_n)$, where $(X'_1, Y'_1) \sim P$ is a test point. Thus, by the tower law,

$$\mathbb{E}[F(\hat{q}_n)] = \mathbb{E}[\mathbb{P}(Y'_1 \in \mathcal{C}_{\hat{q}_n}(X'_1) \mid \hat{q}_n)] = \mathbb{P}(Y'_1 \in \mathcal{C}_{\hat{q}_n}(X'_1)) \geq 1 - \alpha,$$

where the last step holds by the distribution-free marginal coverage guarantee for the split conformal prediction method. This completes the proof for the finite-sample case.

Next, assume $F(q)$ is continuous at $q = q^*$, and that q^* is the unique $(1 - \alpha)$ -quantile, then for any fixed $m \geq 1$. By Theorem 5.6, it holds almost surely that $\hat{q}_n \rightarrow q^*$. Since F is assumed to be continuous at q^* , we then have $F(\hat{q}_n) \rightarrow F(q^*) = 1 - \alpha$, almost surely. By the dominated convergence theorem, then, it holds that $\mathbb{E}[F(\hat{q}_n)^m] \rightarrow (1 - \alpha)^m$, which proves the desired result for the asymptotic setting.

$m \rightarrow \infty$ " = " $\rightarrow$

This proposition reveals a fundamental fact about FWER control with conformal prediction: it is not possible to do much better than a union bound. That is, the aim of FWER control, as stated in (10.7), will only hold (asymptotically) if we choose

$$1 - \alpha_{\text{FWER}} = (1 - \alpha)^m \iff \alpha = 1 - (1 - \alpha_{\text{FWER}})^{1/m} \approx \alpha_{\text{FWER}}/m,$$

i.e., this is nearly the Bonferroni correction. This is because, as the calibration set size n approaches infinity, the events $\{S'_i \leq \hat{q}_n\}_{i \in [m]}$ become approximately independent.

Outlier detection with false discovery rate control

In this section, we turn to a different setting where multiplicity arises: the problem of outlier detection. As before, assume we have i.i.d. calibration points $(X_1, Y_1), \dots, (X_n, Y_n)$ from a distribution P , and an independent set of test points $(X'_1, Y'_1), \dots, (X'_m, Y'_m)$. In contrast to the prediction problem, our goal now is to detect outliers (i.e., test points (X'_i, Y'_i) that were *not* drawn from P), rather than to construct prediction intervals. In particular, $Y'_1, \dots, Y'_m$ are observed, which was not previously the case.

To formalize the question, for each test point $i \in [m]$ define the null hypothesis

$$H_{0,i} : (X'_i, Y'_i) \sim P.$$

Rejecting $H_{0,i}$ means that we claim to have detected that test point (X'_i, Y'_i) is an outlier. For the outlier detection problem, when we are simultaneously testing m many data points, we generally wish to detect outliers without incurring too many false positives. In particular, consider a procedure that sees both the calibration points and the test points and then returns a subset $\mathcal{R} \subseteq \{1, \dots, m\}$ of the test points that are believed to be outliers—this is the set of *rejections* among the null hypotheses $H_{0,1}, \dots, H_{0,m}$ being tested.

When the number of test points m is large, a natural notion of error in this setting is the false discovery rate (FDR), defined as

$$\text{FDR} = \mathbb{E} \left[\frac{\sum_{i=1}^m \mathbb{1} \{i \in \mathcal{R}, H_{0,i} \text{ is true}\}}{\max\{1, |\mathcal{R}|\}} \right],$$

the expected proportion of false rejections.

We now consider a conformal p-value approach towards testing the m null hypotheses with FDR control. Assume that we have a pretrained score function $s : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$. We will test each null hypothesis $H_{0,i}$ by comparing the score $S'_i = s(X'_i, Y'_i)$ against the calibration set scores $S_1, \dots, S_n$. Concretely, we consider the conformal p-value for test point i (recall Section 3.5.1):

$$p_i = \frac{1 + \sum_{j=1}^n \mathbb{1} \{S'_i \leq S_j\}}{n + 1}. \tag{10.8}$$

Notice that the p-values $(p_1, \dots, p_m)$ are dependent, because they are all calculated using the same calibration set. This is the main challenge we address in this section—we will characterize the dependence structure and explain how this enables the use of existing FDR control algorithms together with these conformal p-values.